

Paul Bennet

Curriculum Vitae

Space Telescope Science Institute
3700 San Martin Drive
Baltimore, MD, 21218, USA
✉ pbennet@stsci.edu
🌐 pbennet.github.io

Research Interests

Employment

- Sept. 2024 - Present **STScI Prize Fellowship, NIRCarn, Space Telescope Science Institute, USA**
- Aug. 2020 - Aug. 2024 **Postdoctoral Fellow, Space Telescope Science Institute, USA**
- Summer 2020 **Instructor, Texas Tech University, USA**
- Jan. 2016 – May 2020 **Research Assistant, Texas Tech University, USA**
- Jan. 2014 – Jan. 2016 **Teaching Assistant, Texas Tech University, USA**

Education

- 2017 – 2020 **Ph.D in Physics, Texas Tech University, USA,**
Topic: Low Surface Brightness Dwarf Galaxies in the Big Data Era
- 2014 – 2017 **M.Sc in Physics, Texas Tech University, USA,**
- 2009 – 2013 **M.Phys Physics and Astronomy, University of Southampton, UK,**

Publications

Summary

H-index: 18

1092 Total Citations

Nine Refereed First Author Papers

Thirty Four Refereed co-author Papers

Highlights

1. Bennet et al. 2025, The Orbits of Isolated Dwarfs in the Local Group from New 3D Kinematics: Constraints on First Infall, Backsplash, and Quenching Mechanisms, *The Astrophysical Journal*, 993, 228
 - The most distant proper motion in the literature, for Aquarius dIrr at ~ 1.1 Mpc
 - Explored the links between star formation and orbital histories for a number of distant Local Group dwarf galaxies
2. Bennet et al. 2019, The M101 Satellite Luminosity Function and the Halo to Halo Scatter Among Milky Way Analogues, *The Astrophysical Journal*, 885, 153
 - The first look at a satellite luminosity function into the ultra-faint dwarf regime outside the Local Group

Full Bibliography

Corresponding Author

- **Bennet, Paul**; Patel, Ekta; Sohn, Sangmo Tony; del Pino, Andrés; van der Marel, Roeland P.; Fardal, Mark A.; Spekkens, Kristine; Hunter, Laura Congreve; Besla, Gurtina; Watkins, Laura L.; Weisz, Daniel R.; (2025), The Orbits of Isolated Dwarfs in the Local Group from New 3D Kinematics: Constraints on First Infall, Backsplash, and Quenching Mechanisms, *The Astrophysical Journal*, 993, 228
- **Bennet, Paul**; Patel, Ekta; Sohn, Sangmo Tony; del Pino Molina, Andrés; van der Marel, Roeland P.; Libralato, Mattia; Watkins, Laura L.; Aparicio, Antonio; Besla, Gurtina; Gallart, Carme; Fardal, Mark A.; Monelli, Matteo; Sacchi, Elena; Tollerud, Erik; Weisz, Daniel R. search by orcid; (2024) Proper Motions and Orbits of Distant Local Group Dwarf Galaxies from a combination of *Gaia* and Hubble Data, *The Astrophysical Journal*, 971, 98,
- **Bennet, Paul**; Alfaro-Cuello, Mayte; Pino, Andrés del; Watkins, Laura L.; van der Marel, Roeland P.; Sohn, Sangmo Tony; (2022). Kinematic Structure of the Large Magellanic Cloud Globular Cluster System from *Gaia* eDR3 and Hubble Space Telescope Proper Motions, *The Astrophysical Journal*, 935, 149
- **Bennet, P.**; Sand, D. J.; Crnojević, D.; Weisz, D. R.; Caldwell, N.; Guhathakurta, P.; Hargis, J. R.; Karunakaran, A.; Mutlu-Pakdil, B.; Olszewski, E.; Salzer, J. J.; Seth, A. C.; Simon, J. D.; Spekkens, K.; Stark, D. P.; Strader, J.; Tollerud, E. J.; Toloba, E.; Willamn, B.; (2022). Hubble Space Telescope Imaging of Isolated Local Volume Dwarfs GALFA Dw3 and Dw4, *The Astrophysical Journal*, 924, 98
- **Bennet, P.**; Sand, D. J.; Crnojević, D.; Spekkens, K.; Karunakaran, A.; Zaritsky, D.; Mutlu-Pakdil, B.; (2020). The Satellite Luminosity Function of M101 into the Ultra-faint Dwarf Galaxy Regime , *The Astrophysical Journal*, 893, L9
- **Bennet, P.**; Sand, D. J.; Crnojević, D.; Spekkens, K.; Karunakaran, A.; Zaritsky, D.; Mutlu-Pakdil, B.; (2019). The M101 Satellite Luminosity Function and the Halo to Halo Scatter Among Milky Way Analogues, *The Astrophysical Journal*, 885, 153
- **Bennet, P.**; Sand, D. J.; Zaritsky, D.; Crnojević, D.; Spekkens, K.; Karunakaran, A.; (2018). Evidence for Ultra-diffuse Galaxy 'Formation' through Galaxy Interactions, *The Astrophysical Journal*, 866,L11
- **Bennet, P.**; Sand, D. J.; Crnojević, D.; Spekkens, K.; Zaritsky, D.; Karunakaran, A.; (2017). Discovery of Diffuse Dwarf Galaxy Candidates around M101, *The Astrophysical Journal*, 850:109

Contributing Author

- Ekta Patel, **Paul Bennet**, Sangmo Tony Sohn, Mark A. Fardal, and Roeland P. van der Marel; (2026): Encounters between M33 and Present-day M31 Satellites Hint at a Previous Group Accretion; *The Astrophysical Journal* 999 174
- Eduardo Vitral, Roeland P. van der Marel, Sangmo Tony Sohn, Jorge Peñarrubia, Ekta Patel, Laura L. Watkins, Mattia Libralato, Kevin A. McKinnon, Andrea Bellini, Andrés del Pino, and **Paul Bennet**; (2026); HSTPROMO Internal Proper-motion Kinematics of Dwarf Spheroidal Galaxies. II. Velocity Anisotropy and Dark Matter Cusp Slope of Sculptor; *The Astrophysical Journal* 998 206

- Michael G. Jones, David J. Sand, **Paul Bennet**, Denija Crnojević, Amandine Doliva-Dolinsky, Catherine E. Fielder, Laura C. Hunter, Ananthan Karunakaran, Burçin Mutlu-Pakdil, Deepthi S. Prabhu, Kristine Spekkens, and Dennis Zaritsky; (2026); Star Formation Histories and Gas Content Limits of Three Ultra-faint Dwarfs on the Periphery of M31; *The Astrophysical Journal* 998 142
- Jack T. Warfield, Kevin A. McKinnon, Sangmo Tony Sohn, Nitya Kallivayalil, Alessandro Savino, Roeland P. van der Marel, Andrew B. Pace, Christopher T. Garling, Niusa Ahvazi, **Paul Bennet**, Roger E. Cohen, Matteo Correnti, Mark A. Fardal, Kristen. B. W. McQuinn, Max J. B. Newman, and Eduardo Vitral; (2026); The Proper Motion of Draco II with HST Using Multiple Reference Frames and Methodologies; *The Astrophysical Journal* 998, 3
- Burcin Mutlu-Pakdil, Michael G. Jones, David J. Sand, Denija Crnojević, Kai Herron, Jay Strader, Dennis Zaritsky, **Paul Bennet**, Alex Drlica-Wagner, Quinn O. Casey, Amandine Doliva-Dolinsky, Richard Donnerstein, Catherine E. Fielder, Laura C. Hunter, Annika H. G. Peter, Deepthi S. Prabhu, and Kristine Spekkens; (2026); Hubble Space Telescope Imaging of Three Isolated Faint Dwarf Galaxies beyond the Local Group: Pavo, Corvus A, and Kamino; *The Astrophysical Journal*, 997, 271
- Jones, Michael G.; Rey, Martin P.; Sand, David J.; Spekkens, Kristine; Mutlu-Pakdil, Burçin; Adams, Elizabeth A. K.; **Bennet, Paul**; Crnojević, Denija; Doliva-Dolinsky, Amandine; Donnerstein, Richard; Fielder, Catherine E.; Healy, Julia; Hunter, Laura C.; Karunakaran, Ananthan; Prabhu, Deepthi S.; Zaritsky, Dennis; (2025); Pavo: Stellar Feedback in Action in a Low-mass Dwarf Galaxy; *The Astrophysical Journal*, 990, 164
- Hunter, Laura Congreve; Mutlu-Pakdil, Burcin; Sand, David J.; **Bennet, Paul**; Khim, Donghyeon J.; Crnojević, Denija; Doliva-Dolinsky, Amandine; Durodola, Emmanuel; Fielder, Catherine; Goebel-Bain, Rowan ; Jones, Michael G.; Karunakaran, Ananthan; Spekkens, Kristine; Zaritsky, Dennis; (2025); Identifying Dwarfs of MC Analog Galaxies (ID-MAGE): The Search for Satellites around Low-mass Hosts; *The Astrophysical Journal*, 989, 58
- Doliva-Dolinsky, Amandine; Mutlu-Pakdil, Burcin; Crnojević, Denija; Anbajagane, Dhayaa; Carlin, Jeffrey L.; Medoff, Jonah; Sand, David J.; Tollerud, Erik; Lim, Sungsoo; **Bennet, Paul**; Drlica-Wagner, Alex; Fielder, Catherine E.; Hargis, Jonathan R.; Herron, Kai; Hunter, Laura Congreve; Jones, Michael G.; Karunakaran, Ananthan; Peter, Annika H. G.; Romanowsky, Aaron J.; Spekkens, Kristine; Strader, Jay; Willman, Beth; Carballo-Bello, Julio A.; Cerny, William; Chaturvedi, Astha; Kallivayalil, Nitya; Martínez-Vázquez, Clara E.; Medina, Gustavo E.; Noël, Noelia E. D.; Pace, Andrew B.; Riley, Alex H.; Sakowska, Joanna D.; Smercina, Adam; Vivas, Kathy; Adamów, Monika; Bom, Clecio R.; Choi, Yumi; Ferguson, Peter S.; Navabi, Mahdiah; Zenteno, Alfredo; Delve Collaboration; (2025), The NGC 3109 Satellite System: The First Systematic Resolved Search for Dwarf Galaxies Around an SMC-mass Host, *The Astrophysical Journal*, 989, 21
- Swapnaneel Dey, Michael G. Jones, David J. Sand, Nicolas Mazziotti, Steven Janowiecki, Gregory R. Zeimann, and **Paul Bennet**; (2025), Citizen Science Identification of Isolated Blue Stellar Systems in the Virgo Cluster, *The Astrophysical Journal*, 983, 1

- Catherine E. Fielder, David J. Sand, Michael G. Jones, Denija Crnojević, Alex Drlica-Wagner, **Paul Bennet**, Jeffrey L. Carlin, William Cerny, Amandine Doliva-Dolinsky, Laura C. Hunter, Ananthan Karunakaran, Guilherme Limberg, Burçin Mutlu-Pakdil, Andrew B. Pace, Sarah Pearson, Adam Smercina, Kristine Spekkens, Tjitske Starkenburg, Jay Strader, Guy S. Stringfellow, Erik Tollerud, Clecio R. Bom, Julio A. Carballo-Bello, Astha Chaturvedi, Yumi Choi, David J. James, Clara E. Martínez-Vázquez, Alexander H. Riley, Joanna Sakowska, and Kathy Vivas; (2025), Streams, Shells, and Substructures in the Accretion-built Stellar Halo of NGC 300, *The Astrophysical Journal Letters*, 982, L41
- David J. Sand, Burçin Mutlu-Pakdil, Michael G. Jones, Ananthan Karunakaran, Jennifer E. Andrews, **Paul Bennet**, Denija Crnojević, Giuseppe Donatiello, Alex Drlica-Wagner, Catherine Fielder; (2024), Three Quenched, Faint Dwarf Galaxies in the Direction of NGC 300: New Probes of Reionization and Internal Feedback, *The Astrophysical Journal*, 977, 1
- Catherine Fielder, Michael G. Jones, David J. Sand, **Paul Bennet**, Denija Crnojević, Ananthan Karunakaran, Burçin Mutlu-Pakdil, and Kristine Spekkens; (2024), All Puffed Up: Exploring Ultra-diffuse Galaxy Origins Through Galaxy Interactions, *The Astrophysical Journal*, 168, 212
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- Jones, M. G.; Sand, D. J.; Mutlu-Pakdil, B.; Fielder, C. E.; **Bennet, P.**; Crnojević, D.; Spekkens, K.; Donnerstein, R.; Doliva-Dolinsky, A.; Karunakaran, A.; Strader, J.; Zaritsky, D.; (2024) Corvus A: A Low-mass, Isolated Galaxy at 3.5 Mpc, *The Astrophysical Journal Letters*, 971, L37
- Jones, M. G.; Janowiecki, S.; Dey S.; Sand, D. J.; **Bennet, P.**; Crnojević, D.; Fielder, C. E. ; Karunakaran, A.; Kent, B. R.; Mazziotti, N.; Mutlu-Pakdil, B.; and Spekkens, K.; (2024) Dark No More: The Low-luminosity Stellar Counterpart of a Dark Cloud in the Virgo Cluster , *The Astrophysical Journal Letters*, 966, L15
- Mutlu-Pakdil, B.; Sand, D. J.; Crnojević, D; **Bennet, P.**; Jones, M. G.; Karunakaran, A.; Spekkens, K.; Zaritsky, D.; Caldwell, N.; Fielder, C. E.; Guhathakurta, P.; Seth, A.; Simon, J. D.; Strader, J.; and Toloba, E; (2024) The Faint Satellite System of NGC 253: Insights into Low-density Environments and No Satellite Plane, *The Astrophysical Journal*, 66, 188
- Jones, M. G.; Sand, D. J.; Karunakaran, A.; Spekkens, K.; Oman, K.A.; **Bennet, P.**; Besla, G.; Crnojević, D; Cuillandre, J.; Fielder, C. E. ; Gwyn, S.; Mutlu-Pakdil, B.; (2024), Gas and star formation in satellites of Milky Way analogs, *The Astrophysical Journal* 66, 93
- Watkins, L.L.; van der Marel, R.; **Bennet, P.**; (2024) The Mass of the Large Magellanic Cloud from the Three-Dimensional Kinematics of its Globular Clusters, *The Astrophysical Journal*, 963, 84
- Jones, M. G.; Mutlu-Pakdil, B.; Sand, D. J.; Donnerstein, R.; Crnojević, D; **Bennet, P.**; Fielder, C. E. ; Karunakaran, A.; Spekkens, K.; Strader, J.; Urquhart, R.; Zaritsky, D.; (2023) Pavo: Discovery of a star-forming dwarf galaxy just outside the Local Group, *The Astrophysical Journal Letters*, 957, L5
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- Karunakaran, A.; Spekkens, K.; Caroll R.; Sand, D. J.; **Bennet, P.**; Crnojević, D.; Jones, M. G.; Mutlu-Pakdil, B.; (2022) HI properties of satellite galaxies around local volume hosts, *The Monthly Notices of the Royal Astronomical Society*, 516, 1741–1751
- Sand, D. J.; Mutlu-Pakdil, B.; Jones, M. G.; Karunakaran, A.; Wang, F.; Yang, J.; Chiti, A.; **Bennet, P.**, Crnojević, D, Spekkens, K; (2022). Tucana B: A Potentially Isolated and Quenched Ultra-faint Dwarf Galaxy at $D = 1.4$ Mpc, *The Astrophysical Journal*, 935, L17
- Jones, M. G.; Sand, D. J.; Bellazzini, M.; Spekkens, K.; Karunakaran, A.; Adams, E. A. K.; Battaglia, G.; Beccari, G.; **Bennet, P.**; Crnojević, D.; Caldwell, N.; Fuson, J.; Guhathakurta, P.; Haynes, M. P.; Inoue, J. L.; Magrini, L.; Muñoz, R. R.; Mutlu-Pakdil, B.; Seth, A.; Strader, J.; Toloba, E.; Zaritsky, D.; Young, Blue, and Isolated Stellar Systems in the Virgo Cluster. II. A New Class of Stellar System (2022) *The Astrophysical Journal*, 935, 51
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- Jones, M. G.; **Bennet, P.**; Mutlu-Pakdil, B.; Sand, D. J.; Spekkens, K.; Crnojević, D.; Karunakaran, A.; Zaritsky, D.; (2021). Evidence for Ultra-diffuse Galaxy Formation through Tidal Heating of Normal Dwarfs, *The Astronomical Journal*, 919, 72
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Accepted Proposals Telescope Time/Grants, Optical/IR

P.I.

- 2025 Hubble Space Telescope Cycle 33, 48 primary orbits + 5 parallel orbits (\$138k): Proper Motions of the M31 Satellites: The missing pieces of the 6D puzzle
- 2025 Hubble Space Telescope Cycle 33, 8 primary orbits + 8 parallel orbits (\$49k): The Mysterious Death of a Lonely Dwarf: Who Quenched Cetus?
- 2024 Hubble Space Telescope Cycle 32, Bridging Program, 12 primary orbits (\$25k): 20 Years of time baseline, The 3D kinematics of Centaurus A
- 2024 Hubble Space Telescope Cycle 32, Bridging Program, 6 primary orbits + 6 parallel orbits (\$75k), Joint with JWST (2.4 Primary Hours+ 2.4 parallel hours): Enabling cross instrument proper motions with Draco dSph and NGC 2419
- 2023 Hubble Space Telescope Cycle 31, 42 primary orbits (\$135k): Proper Motions of Galaxies in the M81 Group: Unleashing the Full Power of HST's 20-year Time Baseline
- 2023 Hubble Space Telescope Cycle 31, 16 primary orbits (\$105k): Lone Lion or Part of a Pride: Proper Motion and Orbit of Leo P
- 2022 Hubble Space Telescope Cycle 30, 28 primary orbits (\$228k): New Kids on the Block? Proper Motions of First Infall Galaxies in the Local Group
- 2019 Hubble Space Telescope Cycle 27, 4 primary orbits + 4 parallel orbits (\$49k): The Satellite Luminosity Function of M101 into the Ultra-Faint Dwarf Galaxy Regime

Co.I.

- 2026 Roman Space Telescope Cycle 1, archival (P.I. B. Mutlu-Pakdil): A Rapid Path to High-Impact Early Science: Roman Characterization of Faint Local Volume Dwarf Candidates
- 2026 Roman Space Telescope Cycle 1, (P.I. K. Gilbert): A Legacy Survey of Andromeda and Triangulum with Roman
- 2026 James Webb Space Telescope Cycle 5, Archival (P.I. Mckinnon, K.): The Faint Frontier: Pushing Proper Motion Limits with JWST+Gaia (\$200k)
- 2026 James Webb Space Telescope Cycle 5, 26.0 Primary Hours (P.I. Jones, M. G.): Blue oases in a hostile red desert
- 2026 James Webb Space Telescope Cycle 5, 10.1 Primary Hours (P.I. Hai, T.): The Ancient Star Formation History of a Rare Isolated Ultra-faint Dwarf Galaxy
- 2025 Hubble Space Telescope Cycle 33, 149 primary orbits (P.I. Jones, M. G.): Hubble Census of Nearby Satellites
- 2025 Hubble Space Telescope Cycle 33, 64 primary orbits + 64 parallel orbits (P.I. Kallivayalil, N.): Dynamically Mapping the Satellite Galaxies in the Outer Halo of the Milky Way

- 2025 Hubble Space Telescope Cycle 33, 24 primary orbits + 24 parallel orbits (P.I. Jones, M. G.): Star formation history of the lowest baryonic mass star-forming galaxy known in isolation
- 2024 James Webb Space Telescope Cycle 3, 26.13 Primary Hours (P.I. David J. Sand, \$8k): The Ancient Star Formation History of the Isolated, Quenched Ultra-faint Dwarf Galaxy Tucana B
- 2023 Hubble Space Telescope Cycle 31, 1 primary orbits (P.I. Jones, M. G.): A new gas-rich local dwarf galaxy identified with machine learning
- 2023 Hubble Space Telescope Cycle 31, 16 primary orbits (P.I. Vitral, E.): Shedding Light on Dark Matter: Internal Proper Motions in Ursa Minor
- 2023 Hubble Space Telescope Cycle 31, 3 primary orbits + 3 parallel orbits (P.I. Mutlu-Pakdil, B): Three new isolated, faint and star-forming dwarf galaxies beyond the Local Group
- 2023 Hubble Space Telescope Cycle 31, 3 primary orbits + 3 parallel orbits (P.I. Sand, D. J.): The Ultra-faint Dwarf Galaxy System of NGC300 – A New Frontier and Analog to the Large Magellanic Cloud
- 2023 Hubble Space Telescope Cycle 31, Joint proposal with JVLA, 2 primary orbits + 2 parallel orbits (P.I. Jones, M. G.): Quenching in the lowest mass galaxies at the edge of the Local Group
- 2022 Gemini South 2022B - Fast Turnaround, 4.0 hours, band 1/2, (P.I. Sand, D. J.): A stellar velocity for the isolated, quenched ultra-faint dwarf Tucana B
- 2022 Hubble Space Telescope Cycle 30, 4 primary orbits + 4 parallel orbits (P.I. Mutlu-Pakdil, B): Exploring The Plane of Satellites Around NGC253: A Case Study for Isolated Environments
- 2022 Hubble Space Telescope Cycle 30, Joint proposal with JVLA, 6 primary orbits + 6 parallel orbits (P.I. Sand, D. J.): Ultra-Diffuse Galaxy Formation in Groups – Tidal or Born That Way?
- 2022 Binospec 2022A, 2.1 nights, (P.I. Sand, D. J.): Kinematics of Ultra-Diffuse Galaxy Globular Clusters in the Virgo Cluster
- 2021 Subaru 2021B, 2.1 nights, (P.I. Sand, D. J.): NGC 3109 Ultra-Faint Dwarf Galaxies
- 2021 Canada-France-Hawaii Telescope 2021B, 8 hours, (P.I. Spekkens, K.): How do satellites quench?
- 2021 Hubble Space Telescope Cycle 29, 20 Targets SNAP, (P.I. Jones, M. G.): The globular cluster systems of field ultra-diffuse galaxies
- 2019 Hubble Space Telescope Cycle 27, Joint proposal with JVLA, 4 primary orbits + 4 parallel orbits (P.I. Spekkens, K): Ultra-Diffuse Galaxy Formation through Interactions
- 2019 Hubble Space Telescope Cycle 27, 5 primary orbits (P.I. Mutlu-Pakdil, B): The NGC253 Satellite Luminosity Function: A Case Study for Low Density Environments
- 2019 Gemini North 2019A - Fast Turnaround, 10.0 hours, band 1, (P.I. Sand, D. J.): Kinematics of Ultra-Diffuse Galaxy Globular Clusters in the Virgo Cluster: A Pilot Project
- 2019 Binospec 2019A, 1.5 nights, (P.I. Sand, D. J.): Kinematics of Ultra-Diffuse Galaxy Globular Clusters in the Virgo Cluster
- 2016 Hubble Space Telescope Cycle 24, 19 primary orbits + 19 parallel orbits (P.I. Crnojević, D): An extremely asymmetric dwarf satellite population around M101

- 2016 Hubble Space Telescope Cycle 24, 2 primary orbits (P.I. Sand, D. J.): Two New Local Volume Dwarfs Associated with Compact High Velocity Clouds: Distance, Structure and Star Formation History
- 2016 Gemini North 2016A, 5.6 hours, band 1/2, (P.I. Sand, D. J.): Confirmation of Isolated Low Luminosity Local Volume Galaxies

Accepted Proposals Telescope Time/Grants, Radio Co.I.

- 2025 Jansky Very Large Array 2026A, 115.60 hours (P.I. Jones, M. G.): Adding Gas to the Story: HI-SAGA Survey of satellites around Milky Way Analogs
- 2025 Jansky Very Large Array 2025B, 48.00 hours (P.I. Spekkens, K.): Mapping UDGs to measure dark matter
- 2023 MeerKAT Telescope DDT, 1.25 hours (P.I. Jones, M. G.): Pavo: An isolated, extremely low mass galaxy experiencing a lull in star formation
- 2023 Robert C. Byrd Green Bank Telescope 2024A, 68.0 hours (P.I. Fielder, C): HI Observations of Dwarf Galaxies at the edge of the Local Group and Beyond
- 2023 Jansky Very Large Array 2024A, 27.0 hours (P.I. Jones, M. G.): Witnessing ram pressure stripping of satellites in Milky Way-like systems
- 2023 Jansky Very Large Array 2024A, 36.0 hours (P.I. Motiwala, K.): Mapping HI in UDGs to constrain dark matter
- 2023 Jansky Very Large Array 2024A, 3.5 hours, Joint with HST (P.I. Jones, M. G.): A new gas-rich local dwarf galaxy identified with machine learning
- 2023 Atacama Large Millimeter/submillimeter Array Cycle 10, 69.3 hours, (P.I. Jones, M. G.): Molecular gas content of novel isolated star forming clumps
- 2023 Jansky Very Large Array 2023B, 2.0 hours (P.I. Jones, M. G.): Corvus I, a new star-forming dwarf at the edge of the Local Group
- 2023 Jansky Very Large Array 2023B, 10.5 hours, Joint with HST (P.I. Jones, M. G.): Quenching in the lowest mass galaxies at the edge of the Local Group
- 2023 Jansky Very Large Array DDT, 9.0 hours (P.I. Fielder, C): The origins of a tidally influenced, UV bright ultra-diffuse galaxy
- 2022 Robert C. Byrd Green Bank Telescope 2022B, 96.5 hours (P.I. Jones, M. G.): Cold gas reservoirs of satellites in nearby Milky Way-like systems
- 2022 Jansky Very Large Array 2022B, 36.0 hours (P.I. Jones, M. G.): Resolving UDGs in HI: Are there gas-rich UDGs without dark matter?
- 2021 Jansky Very Large Array 2022A, 42.75 hours (P.I. Jones, M. G.): Satellite Quenching Around Milky Way-like Hosts
- 2021 Jansky Very Large Array 2022A, 5.0 hours (P.I. Jones, M. G.): HI Imaging of an Actively Quenching Dwarf Galaxy
- 2021 Jansky Very Large Array 2022A, 5.0 hours (P.I. Karunakaran, A.): HI Imaging of an Actively Quenching Dwarf Galaxy
- 2021 Jansky Very Large Array 2022A, 16.5 hours, Joint with HST (P.I. Sand D. J.): Ultra-Diffuse Galaxy Formation in Groups – Tidal or Born That Way?
- 2021 Jansky Very Large Array DDT, 3.75 hours (P.I. Sand D. J.): Constraints on the HI content of a candidate ram pressure dwarf in Virgo

- 2021 Robert C. Byrd Green Bank Telescope 2021A, 60.0 hours (P.I. Karunakaran, A.): Mass dependence of Gas Content in LSB Satellite Galaxies
- 2019 Jansky Very Large Array 2019B, 10.0 hours, Joint with HST (P.I. Spekkens, K.): Ultra-Diffuse Galaxy Formation in Groups – Tidal or Born That Way?
- 2018 Robert C. Byrd Green Bank Telescope 2019A, 21.5 hours (P.I. Karunakaran, A.): The HI Content of a Complete Sample of Blue Low Surface Brightness Galaxies
- 2017 Robert C. Byrd Green Bank Telescope 2017B, 18.0 hours (P.I. Karunakaran, A.): Understanding the distribution of satellite candidates around M101
- 2016 Robert C. Byrd Green Bank Telescope 2017A, 32.5 hours (P.I. Spekkens, K.): Probing the asymmetric satellite population of M101

Technical Support Experience

Space Telescope Science Institute

- 2024-26 JWST, NIRCam
 - Improving astrometric distortion corrections
 - Integrating information from other imaging instruments into NIRCam's astrometric solution
 - Moving subarray positions to improve calibration and instrument response
 - Instrument science reviews for both NIRCam and NIRSPEC
 - JDOX updates for NIRCam Imaging
- 2026 HST, ACS
 - Creating an automated optical artifact detection pipeline
- 2026 Science Policy division
 - Panel Manager for the JWST Cycle 6 TAC

Teaching Experience

Johns Hopkins University: Guest Lecturer

- Exploring the Universe with Space Telescopes - Galaxy Kinematics

Texas Tech University: Instructor

- ASTR 1400: Solar System Astronomy

Texas Tech University: Teaching Assistant

- PHYS 3101: Legacy Modern Physics
- ASTR 1400: Solar System Astronomy
- ASTR 1401: Stellar Astronomy
- PHYS 1403: Physics I (Algebra based)
- PHYS 1404: Physics II (Algebra based)
- PHYS 1408: Physics I (Calc. based)
- PHYS 2401: Physics II (Calc. based)

Invited Talks & Seminars

2026 FAU Erlangen-Nürnberg; Physics Colloquium
 2026 University of Surrey; Astronomy Seminar
 2025 University of Victoria; Special Seminar
 2025 Herzberg Astronomy and Astrophysics Research Centre; Astronomy Seminar
 2024 Johns Hopkins University/Space Telescope Science Institute; Colloquium
 2024 San Angelo State University; SPS Seminar
 2023 University of Toronto; Special Seminar
 2023 Queen's University Canada; Astronomy Seminar
 2023 Dartmouth University; Astro Seminar
 2023 University of Rhode Island; Physics Colloquium
 2023 Space Telescope Science Institute; HotSci Colloquium
 2023 Durham University; Special Seminar
 2022 Maryland Astronomy Center for Theory and Computation; Seminar
 2022 Center for Computational Astrophysics, Flatiron Institute; Colloquia
 2021 University of Florida; Astronomy Seminar
 2021 Space Telescope Science Institute; Science Coffee
 2020 University of Chicago; Near Field Cosmology Seminar
 2019 National Optical Astronomy Observatory; FLASH talk
 2018 Texas Tech University; Graduate Research Talk
 2017 Arecibo Observatory; Colloquium

Leadership

2025-Present Postdoc representative to the Science Mission Office, STScI
 2024-Present Postdoc Peer Mentoring Program *Chair*
 2024 Member of the LOC for 2024 STScI Spring symposium: "Recipes to regulate star formation at all scales: From the nearby Universe to the first galaxies."
 2023-2025 Postdoc representative to the Executive Committee of the Science Staff, STScI
 2022 HotSci colloquium series, *Organizer*
 2022-2023 Meet the Speaker colloquium series, *Organizer*
 2021-2025 STScI & JHU Galaxy/AGN Journal Club, *Organizer*
 2019-2020 Sigma Pi Sigma, *Vice-President, TTU Chapter*
 2016-2018 Graduate Society of Physics, *Student Representation Committee Chair*

Honours & Awards

2019 Outstanding Graduate Student Oral Presentation Award, Texas Section APS
 2019-20 Doctoral Dissertation Completion Fellowship
 2019-20 Bucy Graduate Scholarship in Applied Physics
 2017 TTU Department of Physics and Astronomy Outstanding Masters Student
 2017 TTU Departmental Poster Competition, 3rd Place

Professional Service

- Panel Manager: JWST Cycle 6 GO Program
- Panel Support Scientist: HST Cycle 34 GO Program
- Chambliss competition judge, AAS 245
- Session Chair: AAS 245, Session 368
- Instrument Scientist NIRSPEC: JWST Cycle 5 GO Program
- Instrument Scientist NIRCам: JWST Cycle 5 GO Program
- Instrument Scientist NIRCам: JWST Cycle 4 GO Program
- Panel Support Scientist: JWST Cycle 3 GO Program
- Session Chair: AAS 241, Session 328
- Session Chair: AAS 241, Session 253
- Chambliss competition judge, AAS 241
- Leveller: HST Cycle 30 GO Program
- Panel Support Scientist: HST Cycle 29 GO Program
- Panel Support Scientist: JWST Cycle 1 GO Program
- Refereed manuscripts for *The Astrophysical Journal*
- Refereed manuscripts for *The Astrophysical Journal Letters*
- Refereed manuscripts for *The Monthly Notices of the Royal Astronomical Society*
- Refereed manuscripts for *The Monthly Notices of the Royal Astronomical Society Letters*
- Refereed manuscripts for *Astronomy and Astrophysics*
- Refereed manuscripts for *The National Science Review*

Conferences

- 2026 247th AAS Meeting, *Contributed Talk: "Proper Motions Beyond the Local Group"*
- 2025 IAU Symposium 'The Hidden Beauty of the Galactic Outskirts (IAU 403): *Contributed Talk: "The Links between Orbital and Star formation Histories of Dwarf galaxies in the Local Group"* Withdrawn due to family leave
- 2025 245th AAS Meeting, *Poster: "Proper Motions of Distant Local Group Dwarfs: Constraints on Orbits and Star Formation Quenching"*
- 2023 Galactic Frontiers: Dwarf Galaxies in the Local Volume and Beyond, *Contributed Talk: "Orbits of Local Group dwarf galaxies by combining HST and Gaia"*
- 2023 IAU Symposium 379: Dynamical Masses of Local Group Galaxies, *Poster: "3D Velocities of Local Group dwarf galaxies combining HST and Gaia"*
- 2023 241st AAS Meeting, *Contributed Talk: "Local Group dwarf galaxies with GaiaHub: Combining HST and Gaia"*
- 2022 Linking the Galactic and Extragalactic: stellar dynamics and stellar populations of the Milky Way and its siblings?, *Contributed Talk: "Local Group dwarf galaxies with GaiaHub: Combining HST and Gaia"*
- 2022 Aspen Winter Conference: Illuminating Galaxy Formation with Ancient Globular Star Clusters and their Progenitors, *Contributed Talk: "The Proper Motions of Globular Clusters in the Large Magellanic Cloud"*
- 2022 239th AAS Meeting, *Contributed Talk: "The Proper Motions of Globular Clusters in the Large Magellanic Cloud"* Cancelled due to COVID-19

- 2020 235th AAS Meeting, *Contributed Talk: "The M101 Satellite Luminosity Function And The Halo To Halo Scatter Among Local Volume Hosts"*
- 2019 Texas Section American Physics Society, *Contributed Talk: "The M101 Satellite Luminosity Function And The Halo To Halo Scatter Among Local Volume Hosts"*
- 2019 Small Galaxies, Cosmic Questions, *Poster: "The M101 Satellite Luminosity Function And The Halo To Halo Scatter Among Milky Way Analogues"*
- 2019 IAU Symposium 355: The Realm of the Low Surface Brightness Universe, *Poster: "The M101 Satellite Luminosity Function And The Halo To Halo Scatter Among Milky Way Analogues"*
- 2018 The 34th Annual New Mexico Symposium, *Contributed Talk: "Evidence for Ultra-Diffuse Galaxy 'Formation' Through Galaxy Interactions"*
- 2018 Lorentz Centre Workshop, The Bewildering Nature Of Ultra-Diffuse Galaxies, *Contributed Talk: "Diffuse galaxies in some 150 square degrees of the CFHTLS"*
- 2017 229th AAS Meeting, *Contributed Talk: "A new dwarf detection algorithm applied to M101"*
- 2015 225th AAS Meeting, *Poster: "A Systematic Search for Dwarf Counterparts to Ultra Compact High Velocity Clouds"*

Computer skills

- OS MacOS, Linux, Windows
- Programming Python(Intermediate), SQL(Beginner)
- Other Jupyter
- Scientific DS9
- Typography L^AT_EX, Microsoft Office

Volunteer Service

- 2023-Present **JHU PHA Mentorship Program, Mentor**
Mentor for graduate student to help them achieve their personal and professional goals.
- 2017-2018 **Organized Departmental Poster Competition TTU**
- 2016 – 2020 **South Plains Regional Science Fair, Judge**
- 2016 – 2019 **Mentor Tech, Mentor**
Mentor for incoming freshman to help them achieve their personal and professional goals.
- 2016 – 2018 **Mentor Tech, Physics Tutor**
Volunteer physics tutor for monthly study sessions open to all Mentor Tech participants.

Professional Societies

- Sigma Pi Sigma, Physics National Honor Society
- American Astronomical Society
- American Physical Society

Outreach Highlights

- Space Telescope Science Institute
- 2023 **Presentation to the TriState Astronomers**
Presentation to an amateur astronomy society on Galaxies

- 2022 **Presentation to North Colorado Astronomy Society**
Presentation to an amateur astronomy society on Star Clusters
- 2021 **Public lecture at Eureka Public Library**
Public Lecture on the James Webb Space Telescope
Texas Tech University
- 2019-2020 **Astro Night**
Helped organize and led various hands on demonstrations at a monthly astronomy outreach event for the public at TTU
- 2019 **Public Astronomy Outreach**
Event for public viewing of the total lunar eclipse on campus
- 2018 **Science Made Simple**
Hands-on science demonstrations for the public at the TTU Museum.
- 2012 **SUSF Physoc Outreach Launch I**, Part of the founding committee of the Southampton University Space Flight Society
Helped organize balloon launch carrying science experiments with students from a local school and was in charge of coordinating and updating social media.

Languages

English (Native), German (Limited)

Media

- 2022 "The Enduring Mystery of the Dragonfly 44 Galaxy" Quanta Magazine, External expert
- 2020 "Texas Tech astrophysics researcher has projects approved with Hubble Space Telescope", Lubbock Avalanche-Journal
- 2019 "Physics Doctoral Student Will Use Hubble Space Telescope to Compare Galaxies", Lubbock Avalanche-Journal
- 2019 "Physics Doctoral Student Will Use Hubble Space Telescope to Compare Galaxies" Texas Tech Today, related to the "The Satellite Luminosity Function of M101 into the Ultra-Faint Dwarf Galaxy Regime" HST proposal
- 2019 "Physics Doctoral Student Will Use Hubble Space Telescope to Compare Galaxies" Texas Tech Monthly Newsletter, related to the "The Satellite Luminosity Function of M101 into the Ultra-Faint Dwarf Galaxy Regime" HST proposal

References

Prof. David Sand

Ph.D. Advisor

University of Arizona

933 N Cherry Ave

Tucson, AZ 85719

✉ dsand@email.arizona.edu

Dr. Roeland van der Marel

Former Research Supervisor

Space Telescope Science Institute

3700 San Martin Dr
Baltimore, MD 21218
✉ marel@stsci.edu

Dr. Martha Boyer
Current Functional Manager
Space Telescope Science Institute
3700 San Martin Dr
Baltimore, MD 21218
✉ mboyer@stsci.edu